

Consistent and unbiased estimation of the hypervolume of an unknown true Pareto front

(Resolving uncertainty about the quality of results)

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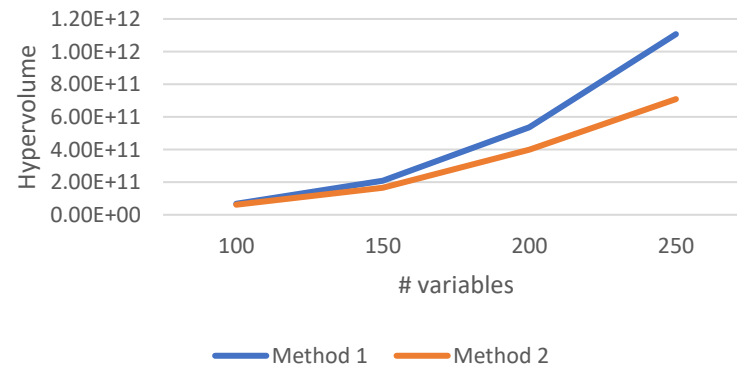
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Motivation

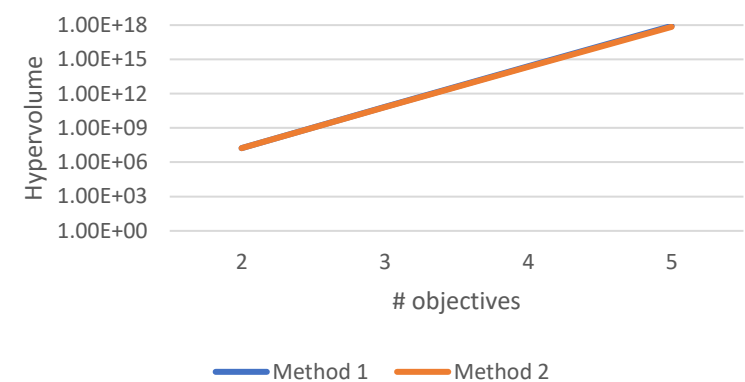
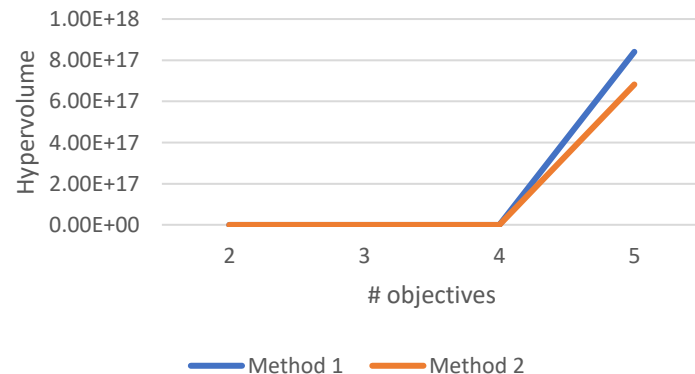
- Set quality indicators play an increasingly important role in (evolutionary) multiobjective optimization, as they are essential for both the **evaluation** and design of (E)MO algorithms
- Hypervolume is most likely the most often used set quality indicator in (E)MO

Results of a (real) computational experiment for multiobjective knapsack problem

Expriment I (d = 3)		
# variables	Method 1	Method 2
100	6.80E+10	6.29E+10
150	2.09E+11	1.66E+11
200	5.36E+11	4.00E+11
250	1.11E+12	7.10E+11



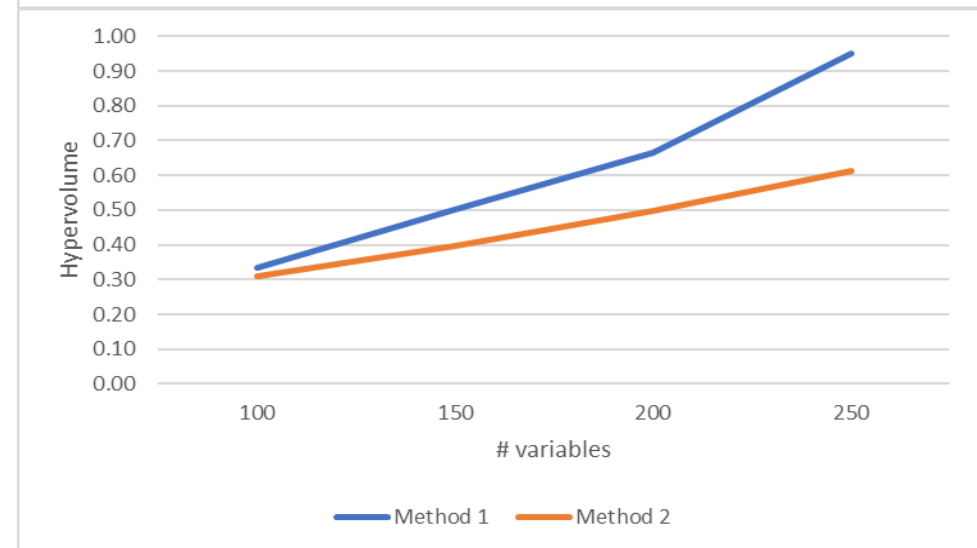
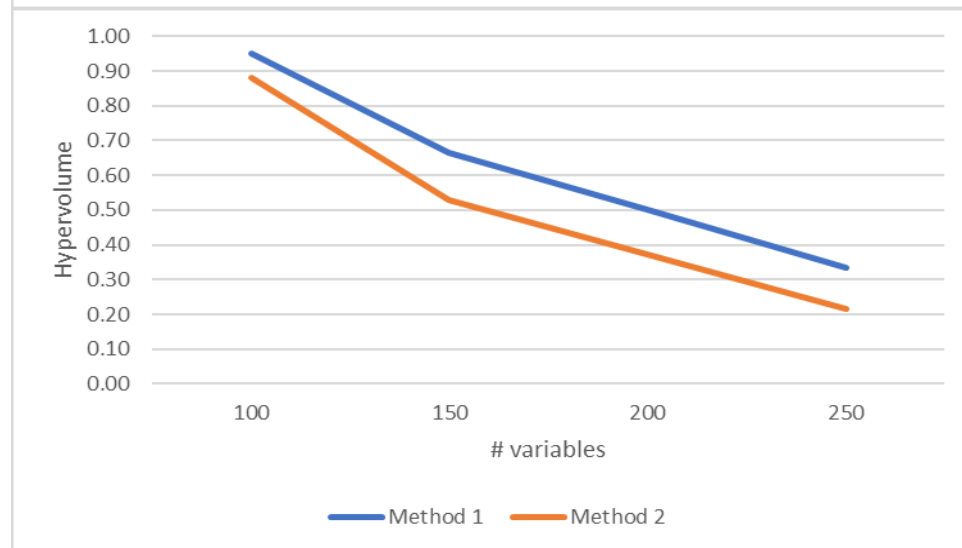
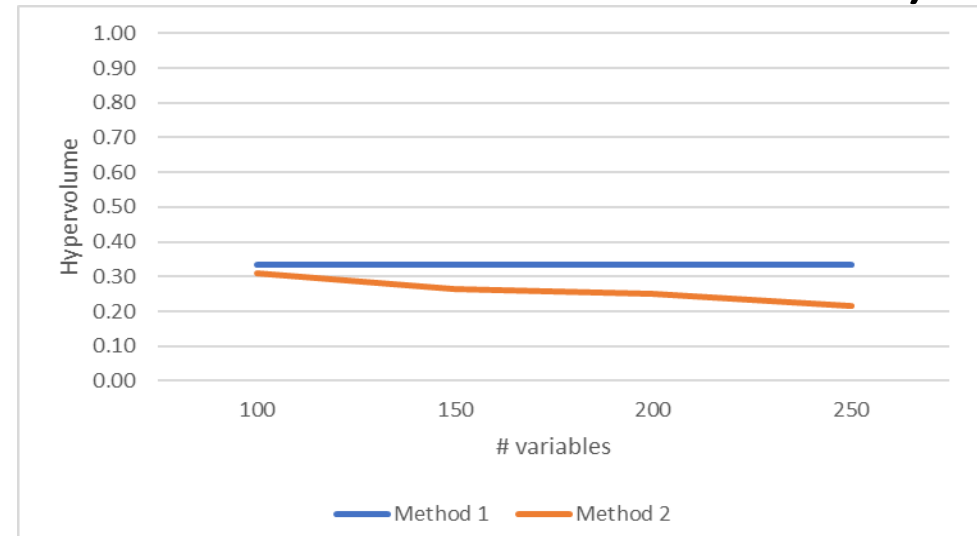
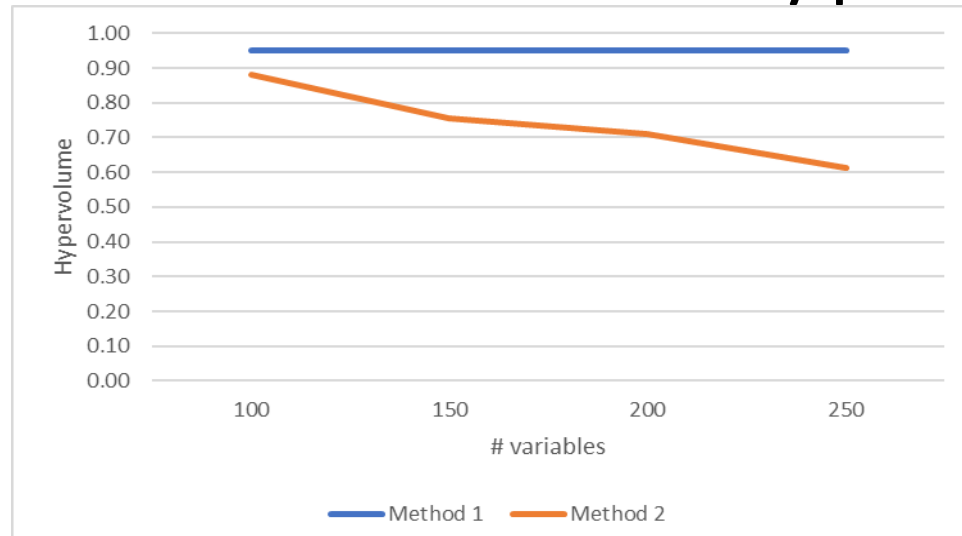
Expriment II (n = 100)		
# objectives	Method 1	Method 2
2	1.73E+07	1.66E+07
3	6.80E+10	6.29E+10
4	2.51E+14	2.20E+14
5	8.41E+17	6.83E+17



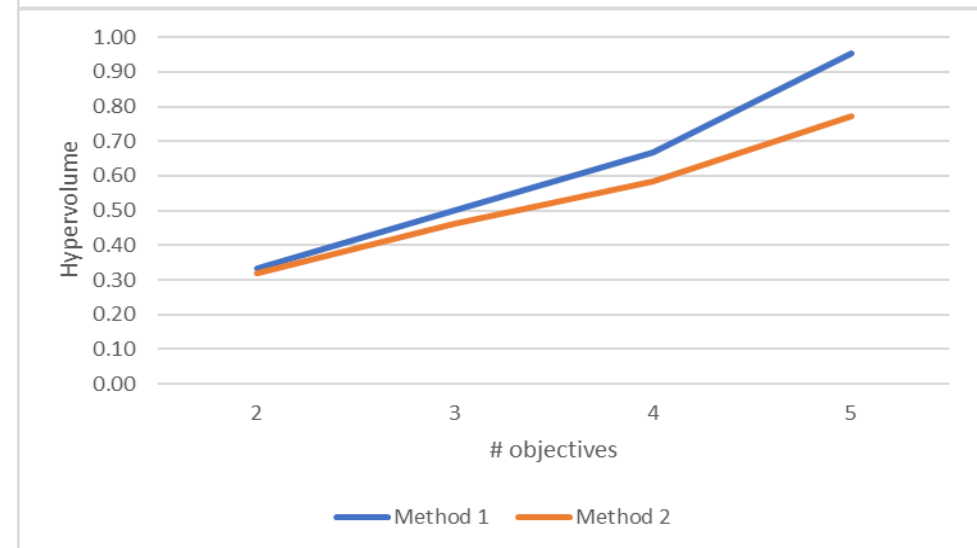
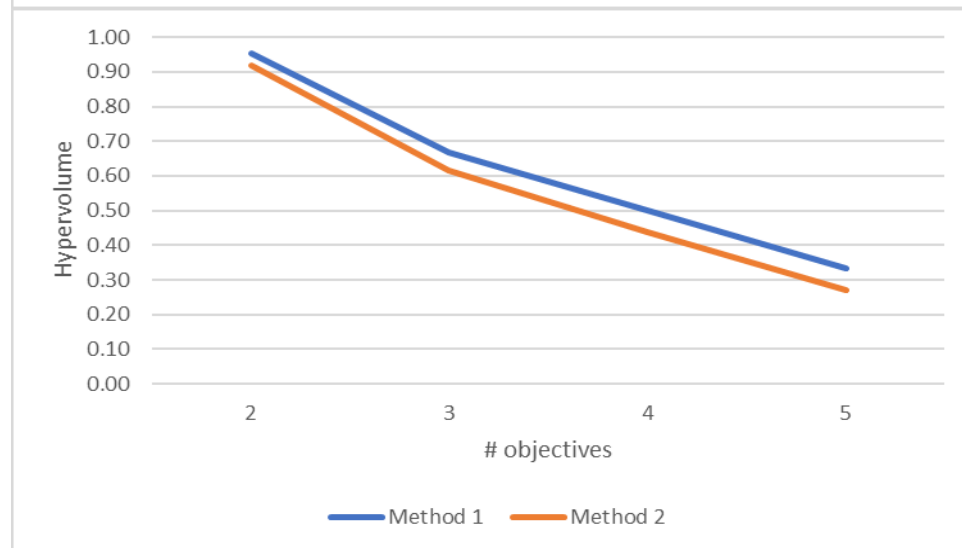
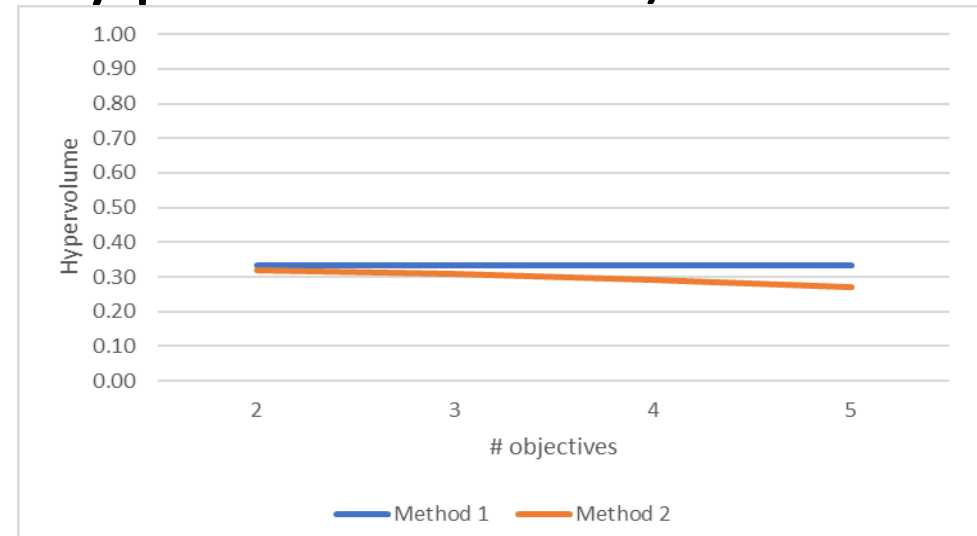
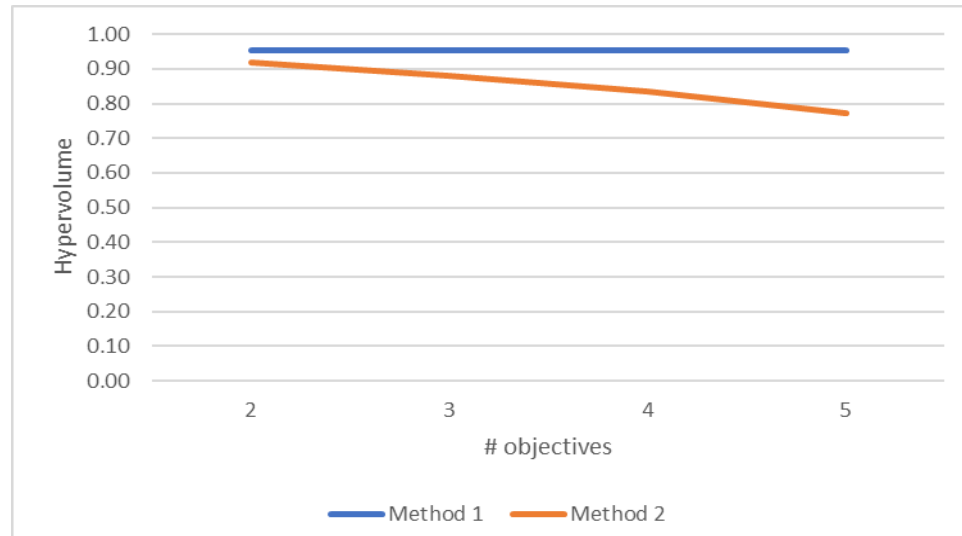
A commonplace in (single objective) optimization

- „If we do not know the gap between the value of objective function for a heuristic solution and the optimum value, we know nothing”

Some potential results of experiment I (when normalized to hypervolume of the true PF)



Some potential results of experiment II (when normalized to the true hypervolume)



How can we know hypervolume of the true PF

- We could generate the true PF (and calculate its hypervolume)
 - See Ehrgott, M., Köksalan, M., Kadziński, M., Deb, K., 2025. Fifty years of multi-objective optimization and decision-making: From mathematical programming to evolutionary computation. EJOR, Section 3, for a recent survey.
- In general, very time consuming
 - For example, Tamby, S., Vanderpooten, D., 2021. Enumeration of the nondominated set of multi-objective discrete optimization problems. INFORMS Journal on Computing 33, 72–85, consider knapsack instances with up to $(d = 2, n = 200)$, $(d = 3, n = 100)$, $(d = 4, n = 50)$, $(d = 5, n = 50)$, $(d = 6, n = 30)$ – running times close or exceeding 1 hour
 - Calculation of the exact hypervolume of a large true PF (even > 8000 above) may also be very time consuming – we anyway may need an estimation of HV

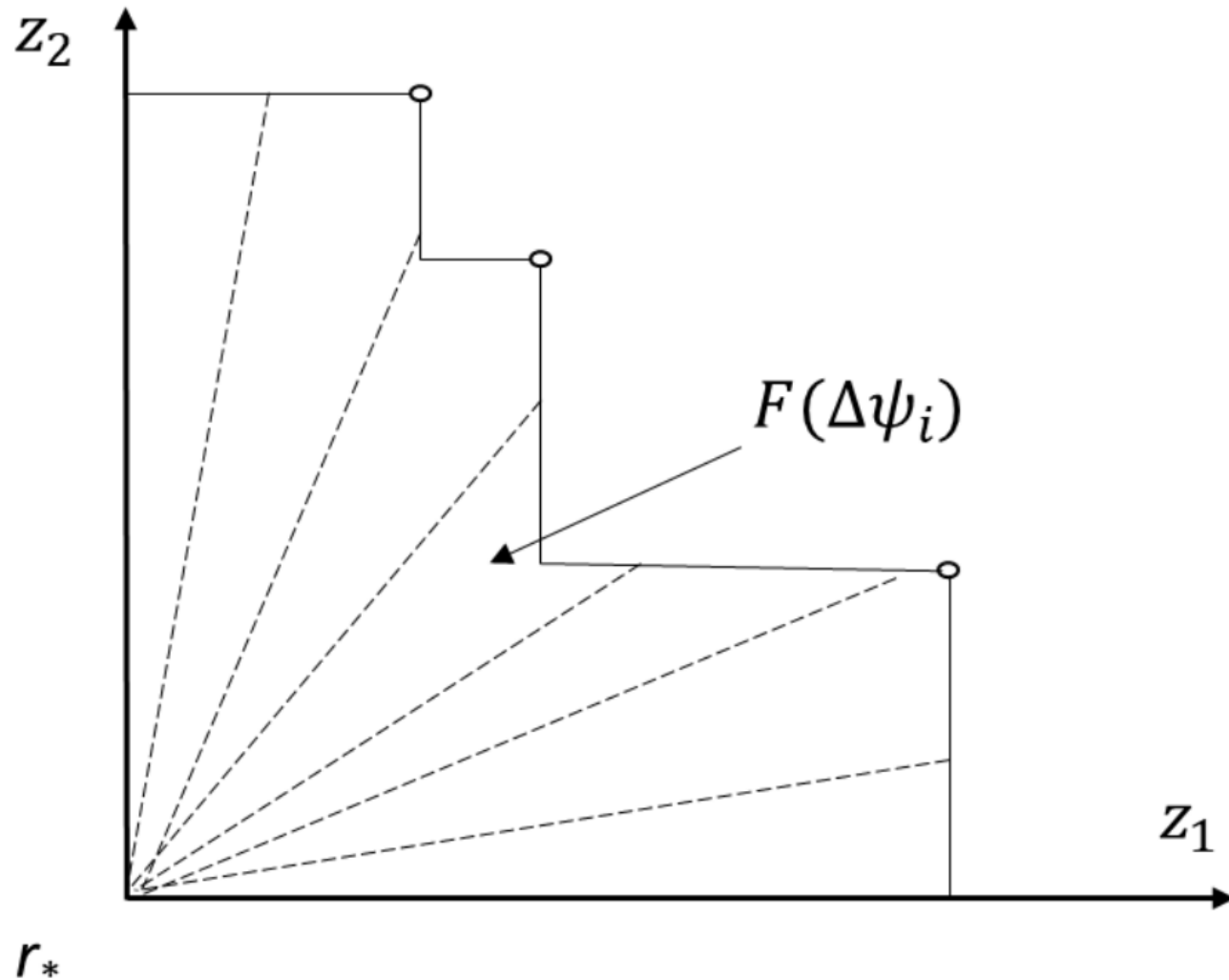
Distance based estimation of hypervolume

- Deng, J., Zhang, Q., 2019. Approximating hypervolume and hypervolume contributions using polar coordinate. IEEE TEVC 23, 913–918.
 - Jaskiewicz, A., Zielniewicz, P., 2025. Improving the efficiency of the distance-based hypervolume estimation using ND-tree. IEEE TEVC 29, 726–733.

$$\begin{aligned} H(P, r_*) &= \frac{1}{d} \int_{\Psi^+} L(P, r_*, \psi)^d d\psi \\ &= \frac{2 \pi^{d/2}}{\Gamma(\frac{d}{2}) 2^d d} E_{\psi \in \Psi^+} (L(P, r_*, \psi)^d) \end{aligned} \tag{8}$$

where $\Psi^+ = \{\psi \in \mathbb{R}^d \mid |\psi| = 1 \wedge \forall j, \psi_j \geq 0\}$ denotes the set of normalized direction vectors, and $L(P, r_*, \psi)$ represents the Euclidean distance from r_* to the upper boundary of the region weakly dominated by P along direction ψ .

Distance based estimation of hypervolume - idea



Euclidean distance from r_* to the attainment surface along direction ψ

- Ma, X., Zhang, Q., Tian, G., Yang, J., Zhu, Z., 2018. On Tchebycheff decomposition approaches for multiobjective evolutionary optimization. IEEE TEVC 22, 226–244.

$$\begin{aligned} L(P, r_*, \psi) &= \max_{z \in S} (s_{\infty}^m(z, r_*, \Lambda(\psi))) \\ &= \max_{z \in S} (\min_j -\lambda_j(\psi)(r_{*j} - z_j)) \end{aligned}$$

$$\text{where } \lambda_j(\psi) = \frac{1}{\psi_j}.$$

Numerical integration

- Expected value is estimated by an average over randomly sampled directions
- Provides consistent and unbiased estimation of the hypervolume of the true PF:

$$\tilde{H}(P, r_*) = \frac{1}{d} \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2}) 2^d} \frac{1}{n} \sum_{i=1}^n L(P, r_*, \psi_i)^d \xrightarrow{n \rightarrow \infty} H(P, r_*)$$

$$E(\tilde{H}(P, r_*)) = \frac{1}{d} \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2}) 2^d} E\left(\frac{1}{n} \sum_{i=1}^n L(P, r_*, \psi_i)^d\right) = H(P, r_*)$$

Confidence interval (Gandibleux, Jaszkievicz)

$$[\tilde{H}(P, r_*) - z^* \frac{s}{\sqrt{n}}, \tilde{H}(P, r_*) + z^* \frac{s}{\sqrt{n}}]$$

$$s = \frac{1}{d} \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2}) 2^d} \sqrt{\frac{1}{n-1} \sum_{i=1}^n (L(P, r_*, \psi_i)^d - \overline{L(P, r_*, \psi_i)^d})^2}$$

Distance based estimation of hypervolume

- Originally proposed for estimating hypervolume of a discrete, explicitly known set of points (to improve time efficiency of exact hypervolume calculation) ...
 - The point optimizing the inverse Chebycheff functions is found by exhaustive search over all points or using ND-Tree
- ... but applies to continuous and/or unknown (implicitly known) set, e.g. true PF of MOOP (to avoid costly generation on the true PF)
- Possibility mentioned (theoretically) in
 - Zhang, X., Lin, X., Xue, B., Chen, Y., Zhang, Q., 2023. Hypervolume maximization: a geometric view of pareto set learning, in: Proceedings of NeurIPS.

Linear and differentiable reformulation

- Direct formulation (non-linear, non-differentiable):

$$\begin{aligned} & \text{maximize} && (\min_j -\lambda_j(\psi)(r_{*j} - z_j(x))) \\ & \text{s.t.} && x \in X \subset \mathbb{R}^n \end{aligned} \tag{15}$$

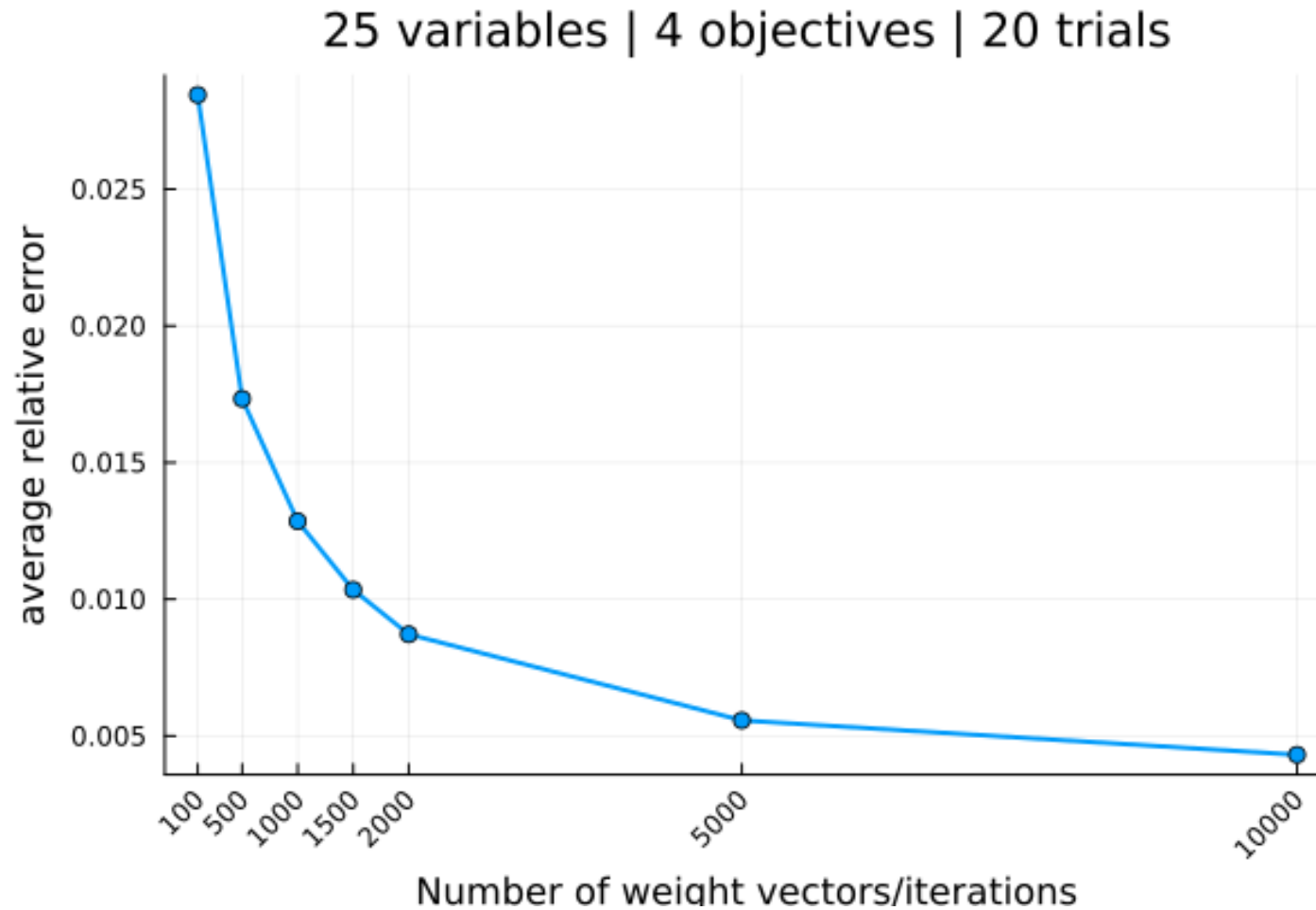
Linear and differentiable reformulation (Gandibleux, Jazskiewicz):

$$\begin{aligned} & \text{maximize} && \alpha \\ & \text{s.t.} && \alpha \leq -\lambda_j(\psi)(r_{*j} - z_j(x)), \quad j = 1, \dots, d \\ & && x \in X \subset \mathbb{R}^n \end{aligned} \tag{16}$$

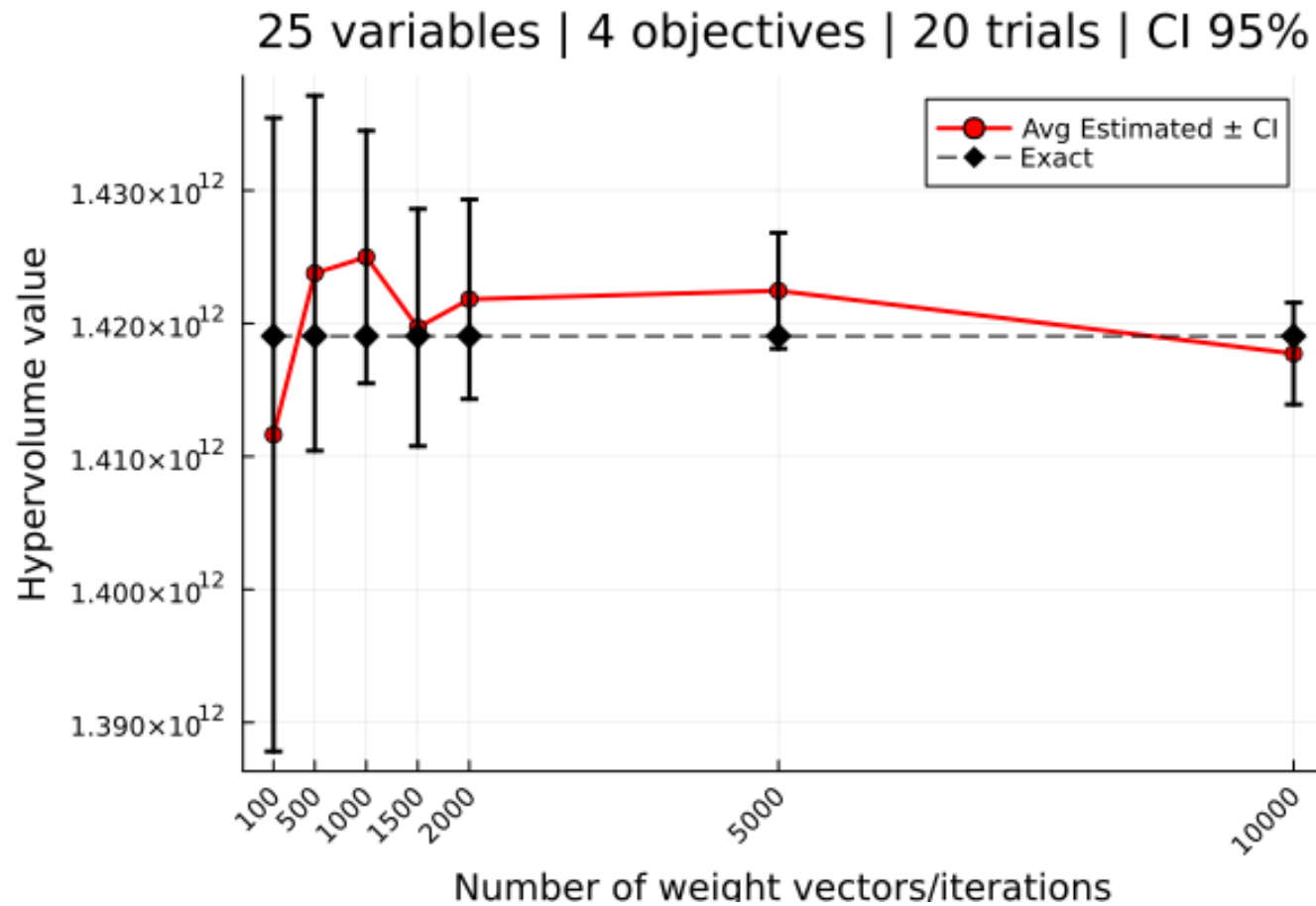
Implementation in Julia programming language (Gandibleux, Jaszkievicz)

- Open source available at <https://github.com/xgandibleux/HypervolumeEstimator>
- General purpose with an example for the knapsack problem
- Tested with GLPK and Gurobi solver (easy use of other solvers)

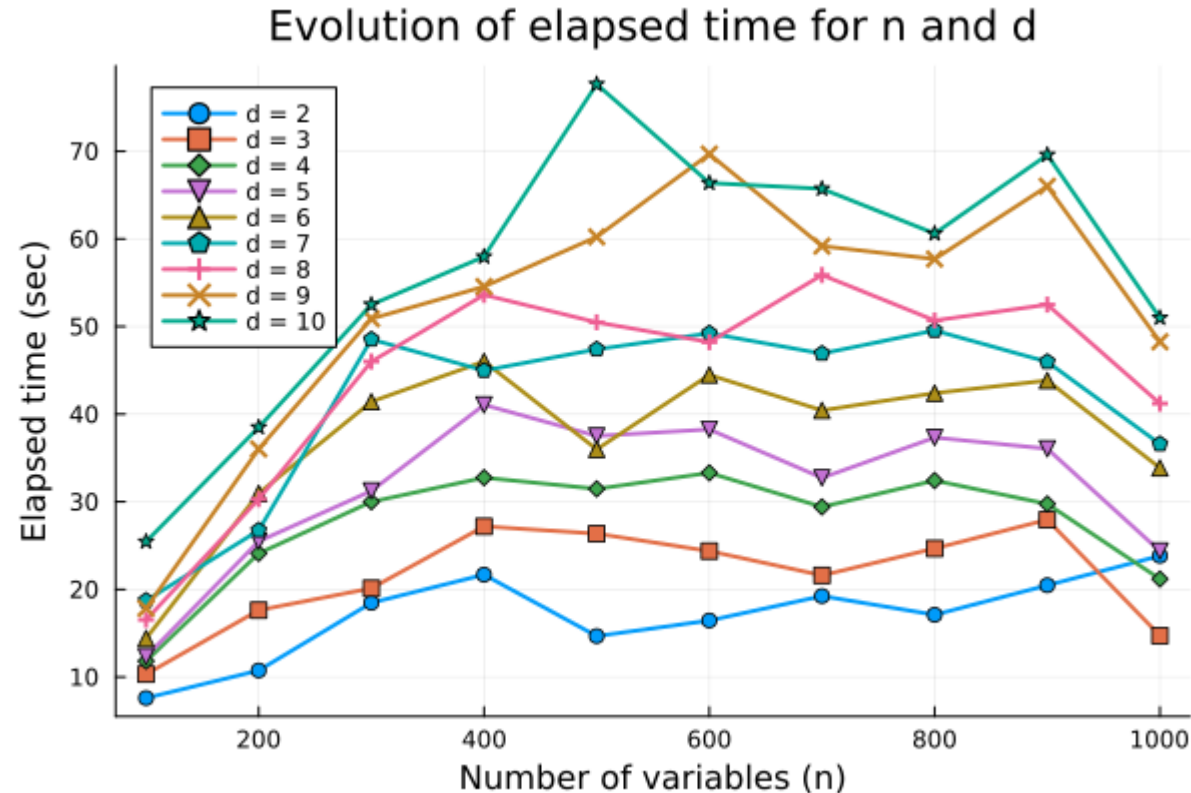
Validity tests



Validity tests



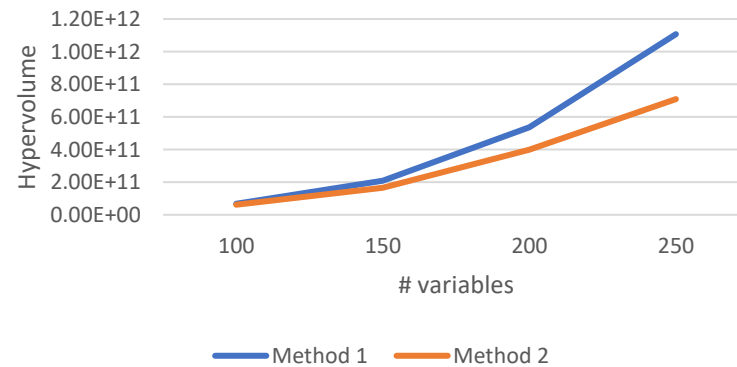
Efficiency with Gurobi Optimizer version 13.0 (2000 weights)



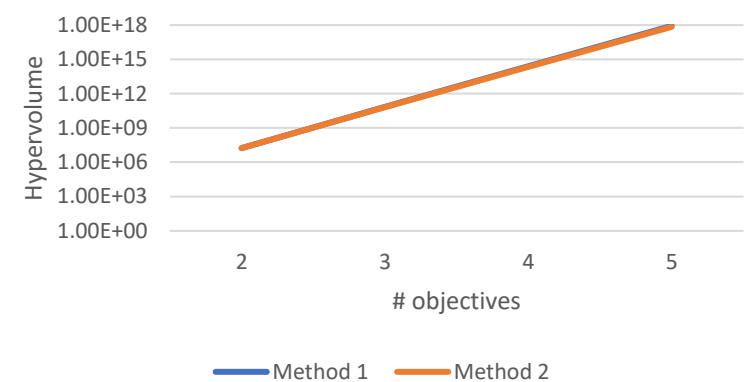
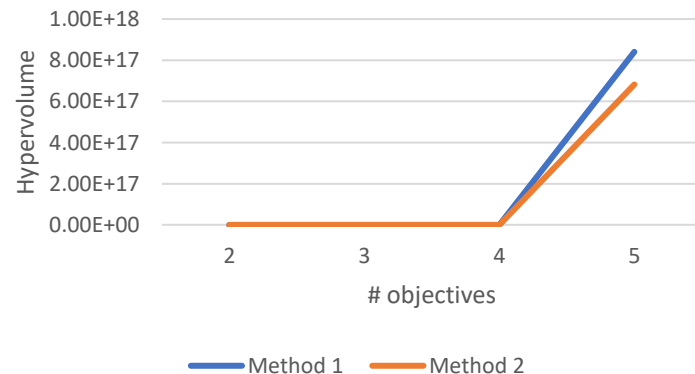
- Reminder: (SOTA) Tamby, S., Vanderpooten, D., 2021. Enumeration of the nondominated set of multiobjective discrete optimization problems. INFORMS Journal on Computing 33, 72–85, consider knapsack instances with up to ($d = 2$, $n = 200$), ($d = 3$, $n = 100$), ($d = 4$, $n = 50$), ($d = 5$, $n = 50$), ($d = 6$, $n = 30$) – running times exceeding 1 hour

Results of a (real) computational experiment for multiobjective knapsack problem

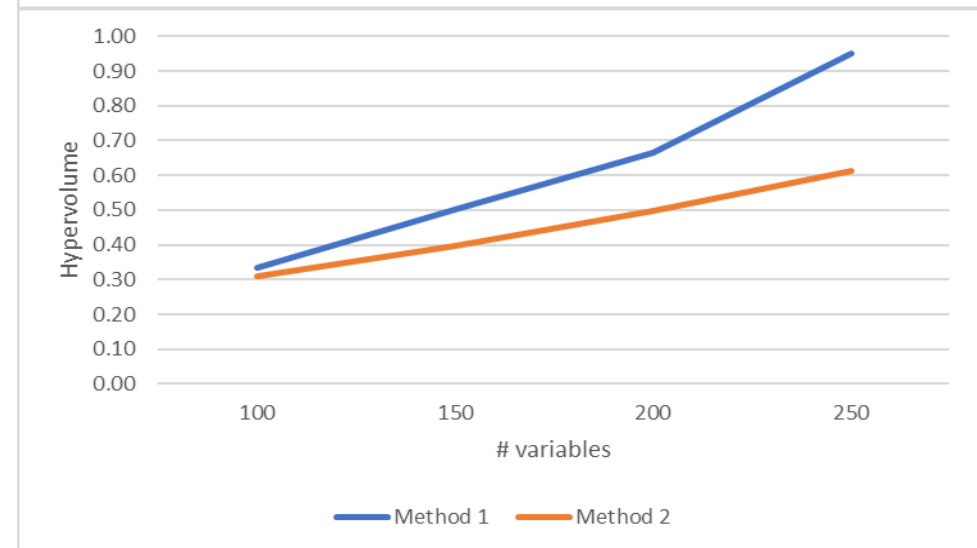
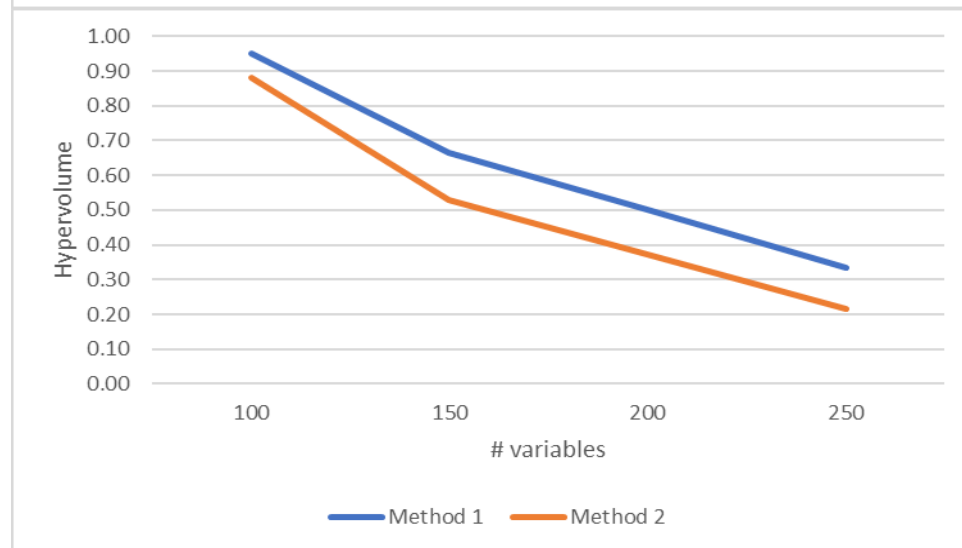
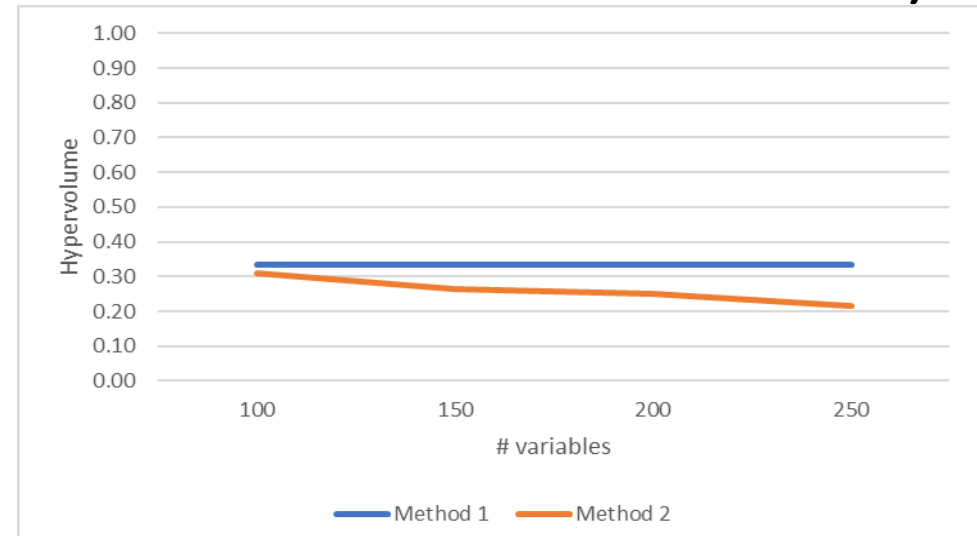
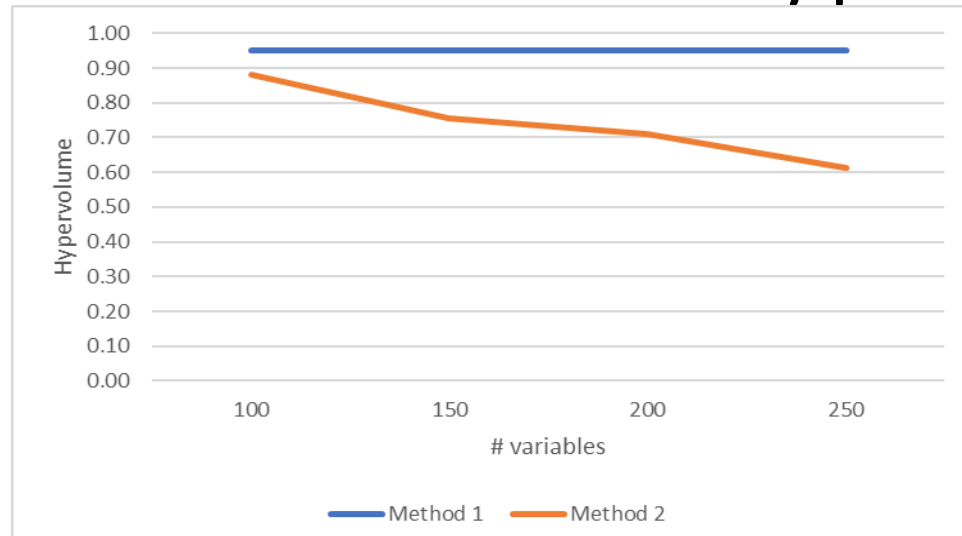
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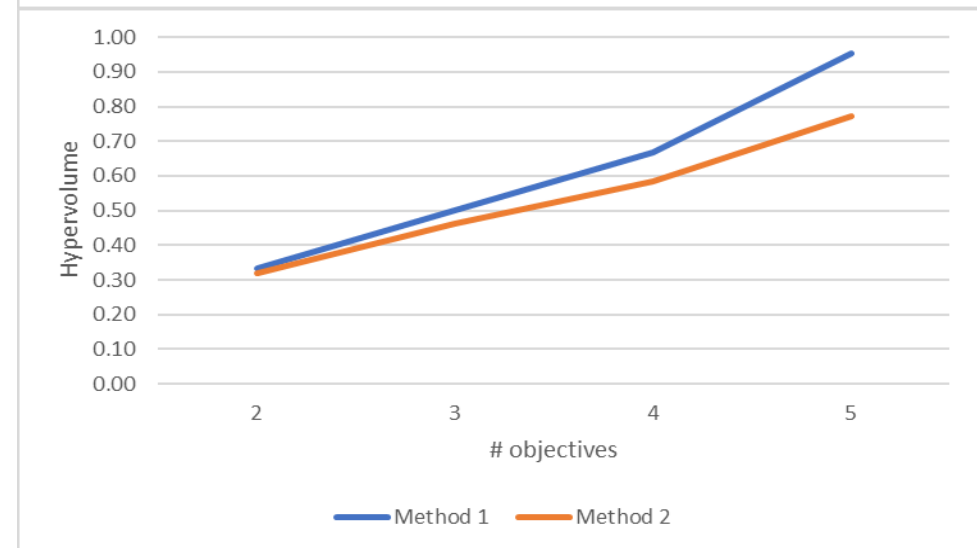
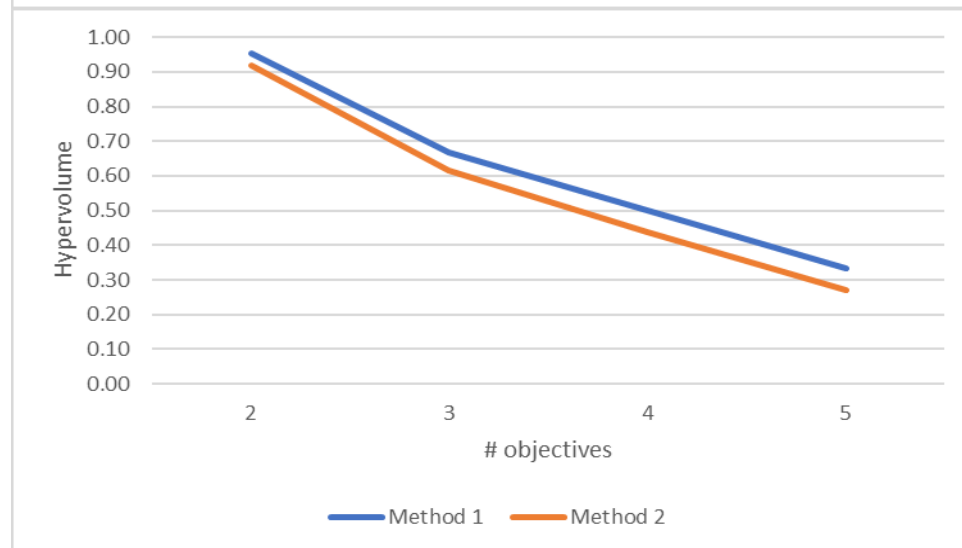
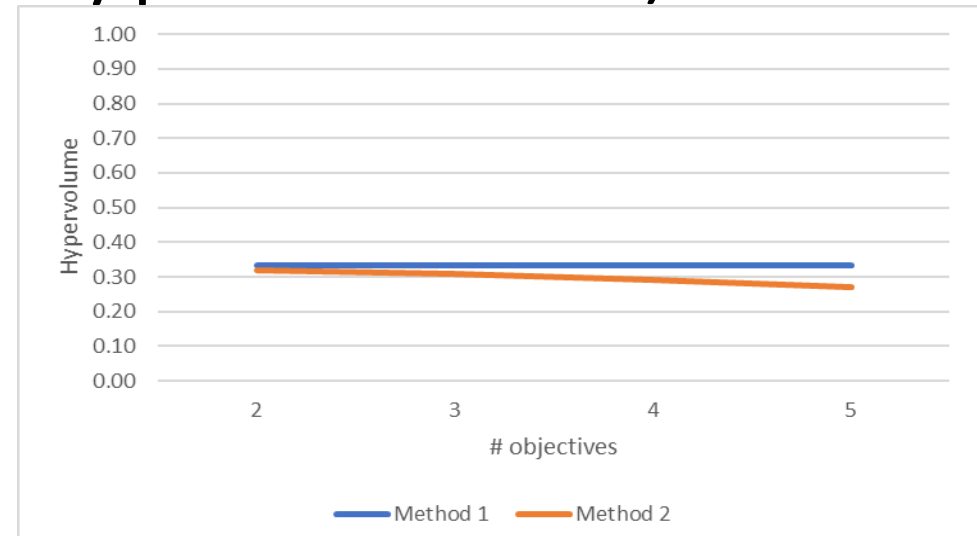
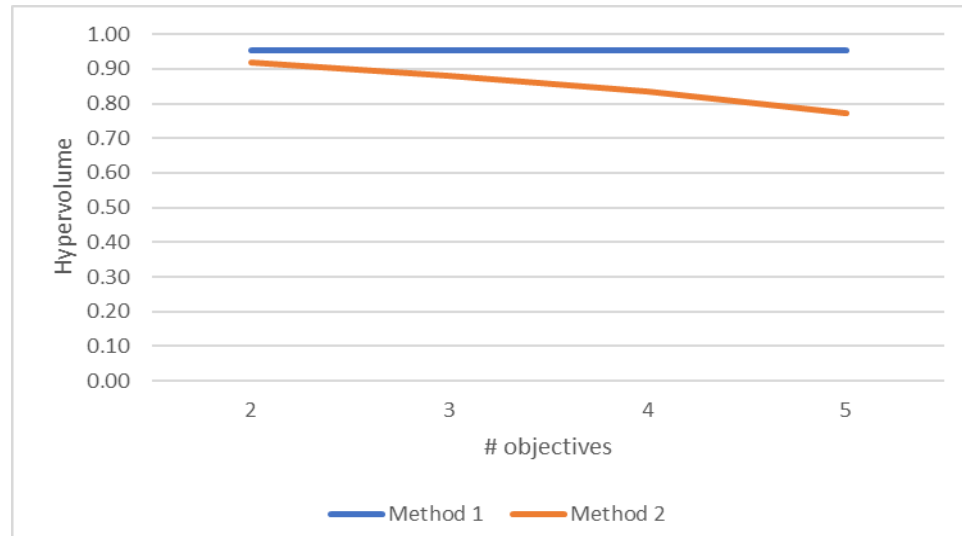
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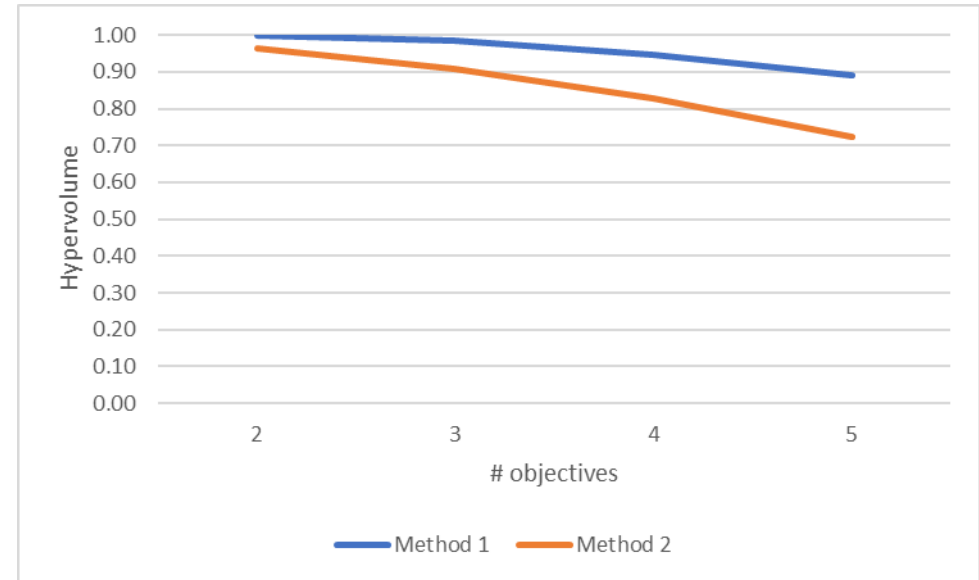
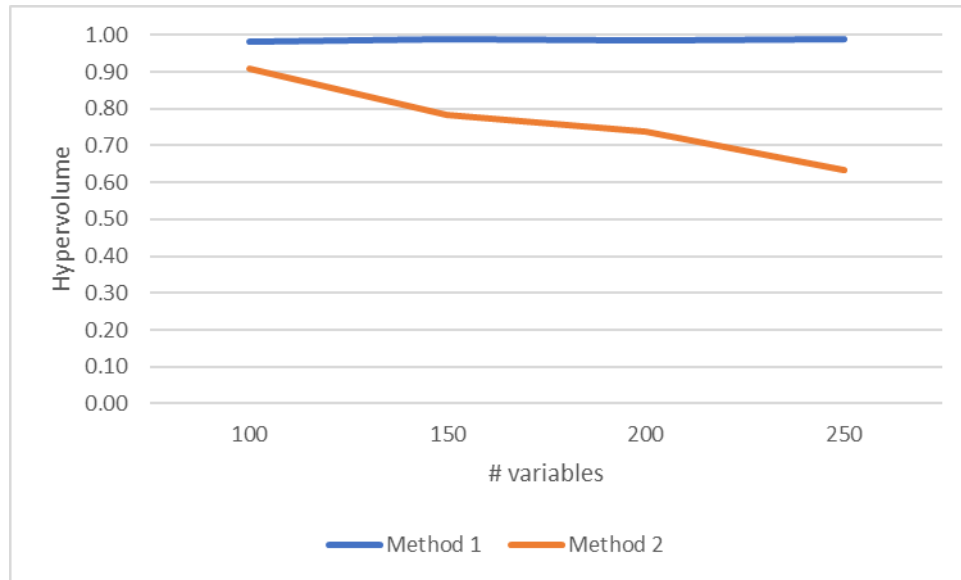
Some potential results of experiment I (when normalized to hypervolume of the true PF)



Some potential results of experiment II (when normalized to the true hypervolume)



Results with normalization to the estimated hypervolume of the true PF



Stochastic case

$L(P, r_*, \psi) = \max_{z \in S} (s_\infty^m(z, r_*, \Lambda(\psi)))$ is a random variable

$$\begin{aligned} E(\tilde{H}(P, r_*)) &= C E\left(\sum_{i=1}^n L(P, r_*, \psi_i)^d\right) = \\ &= C \sum_{i=1}^n E(L(P, r_*, \psi_i)^d) = \\ &= C \sum_{i=1}^n E\left(\left(\max_{z \in S} (s_\infty^m(z, r_*, \Lambda(\psi_i)))\right)^d\right) \end{aligned}$$

- So we need:

$$E \left(\left(\max_{z \in S} \left(s_{\infty}^m(z, r_*, \Lambda(\psi)) \right) \right)^d \right)$$

- Or consider R2 quality indicator which uses linear aggregation and in many cases gives the same results as hypervolume
 - Yang Nan, Hisao Ishibuchi, Tianye Shu, and Ke Shang. 2025. R2 Indicator Analysis using the Optimal Distributions of Solutions for R2 and Other Indicators. GECCO '25

Summary

- Hypervolume of the true Pareto front can be estimated in consistent and unbiased way without generating this set, and the estimation is associated with a confidence interval
- Obtaining such estimation is much more efficient than generation of the true PF
- This estimation may be used to estimate the gap of the approximate results to the best possible hypervolume
- Maybe it is a good approach for estimating hypervolume with stochastic objectives?

- Preprint available at Optimization Online: Xavier Gandibleux, Andrzej Jaszewicz, Consistent and unbiased estimation of the hypervolume of an unknown true Pareto front, <https://optimization-online.org/?p=31635>
- Open source Julia implementation available at <https://github.com/xgandibleux/HypervolumeEstimator>